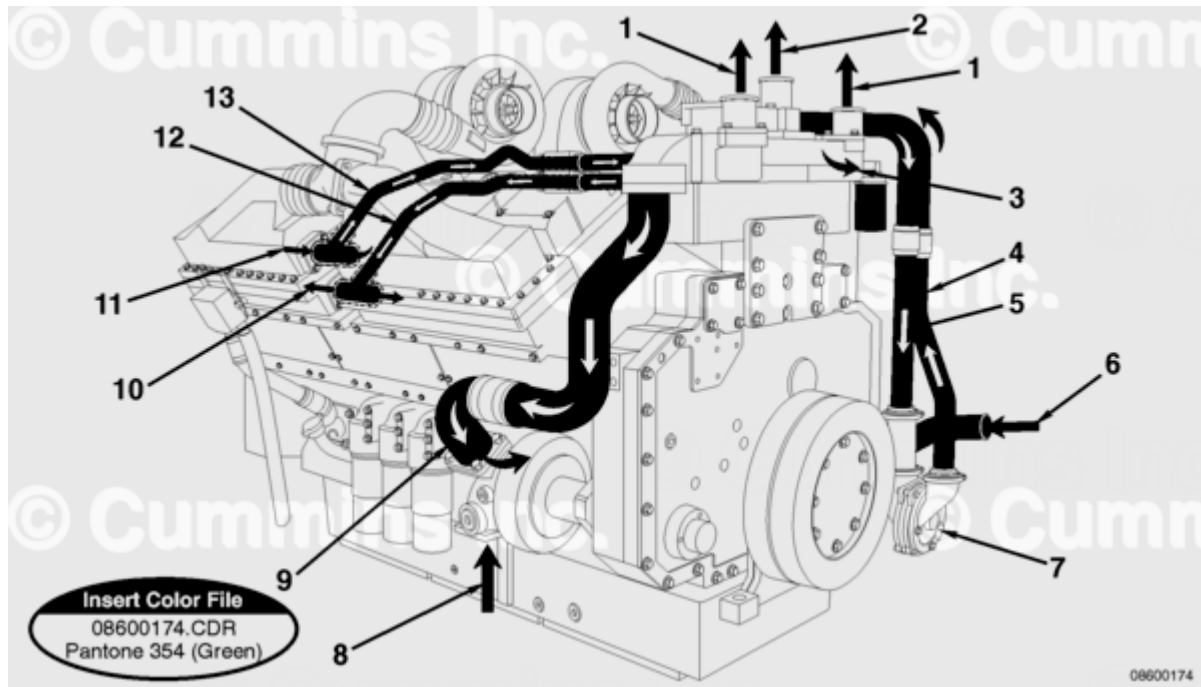


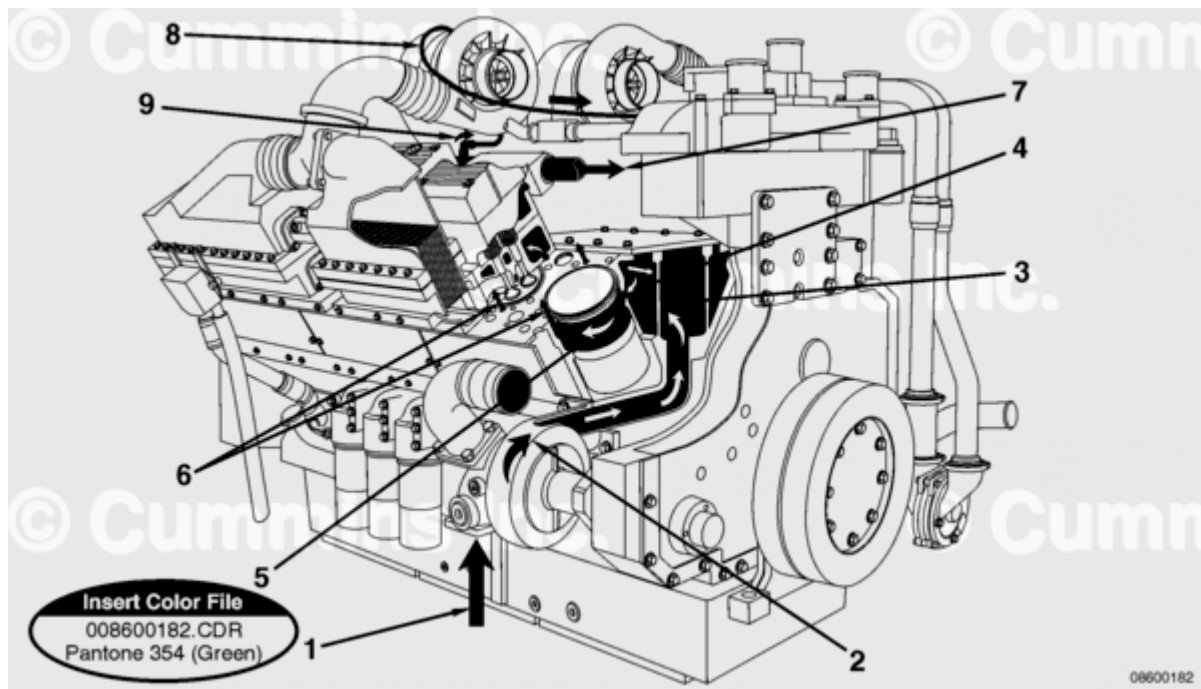
Trainings ▾

Flow Diagram



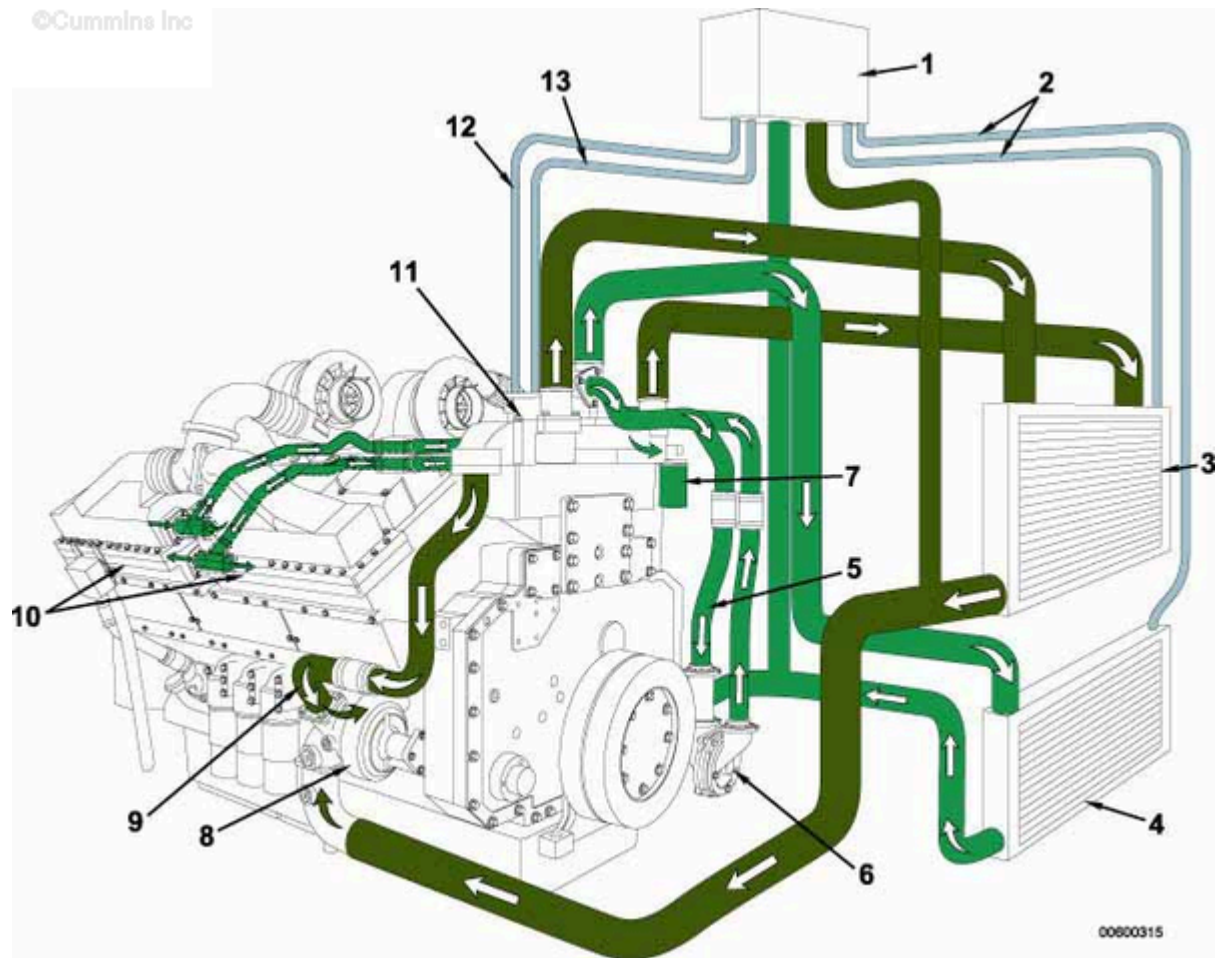
Cooling System (QSK45 engine shown), with Mechanically Actuated Injectors

1. Coolant flow to engine radiator
2. Coolant flow to low temperature aftercooler radiator
3. Coolant flow to coolant filters
4. Coolant flow from low temperature aftercooler pump
5. Coolant bypass to low temperature aftercooler water pump
6. Coolant flow from low temperature aftercooler radiator
7. Low temperature aftercooler water pump
8. Coolant inlet to water pump
9. Coolant bypass return to pump
10. Coolant flow to aftercooler core
11. Coolant flow from aftercooler core
12. Coolant flow to aftercooler from thermostat housing
13. Coolant flow from aftercooler to thermostat housing.



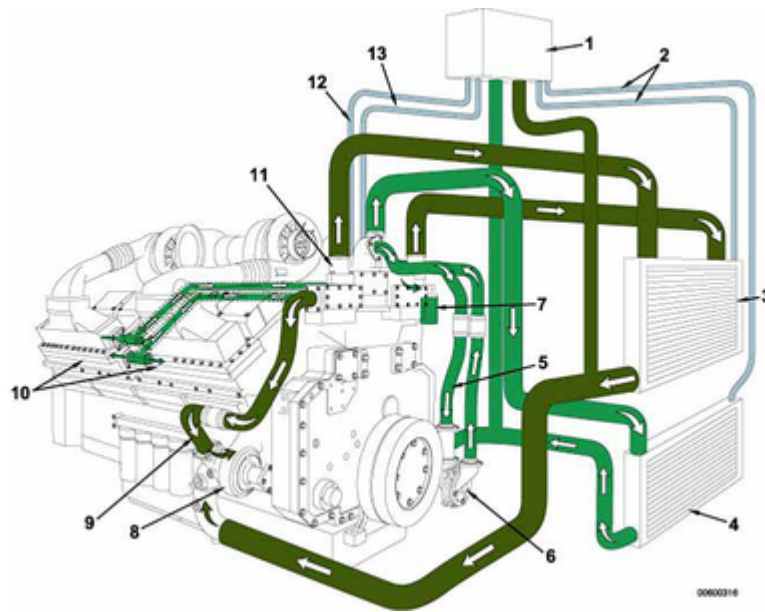
Cooling System (Continued) (QSK45 engine shown), with Mechanically Actuated Injectors

1. Coolant inlet
2. Water pump
3. Coolant flow around oil coolers
4. Coolant flow to cylinder liners
5. Coolant flow around cylinder liners
6. Coolant flow to cylinder head
7. Coolant flow through water manifold to thermostat housing
8. Coolant return from turbocharger
9. Coolant supply to turbocharger.



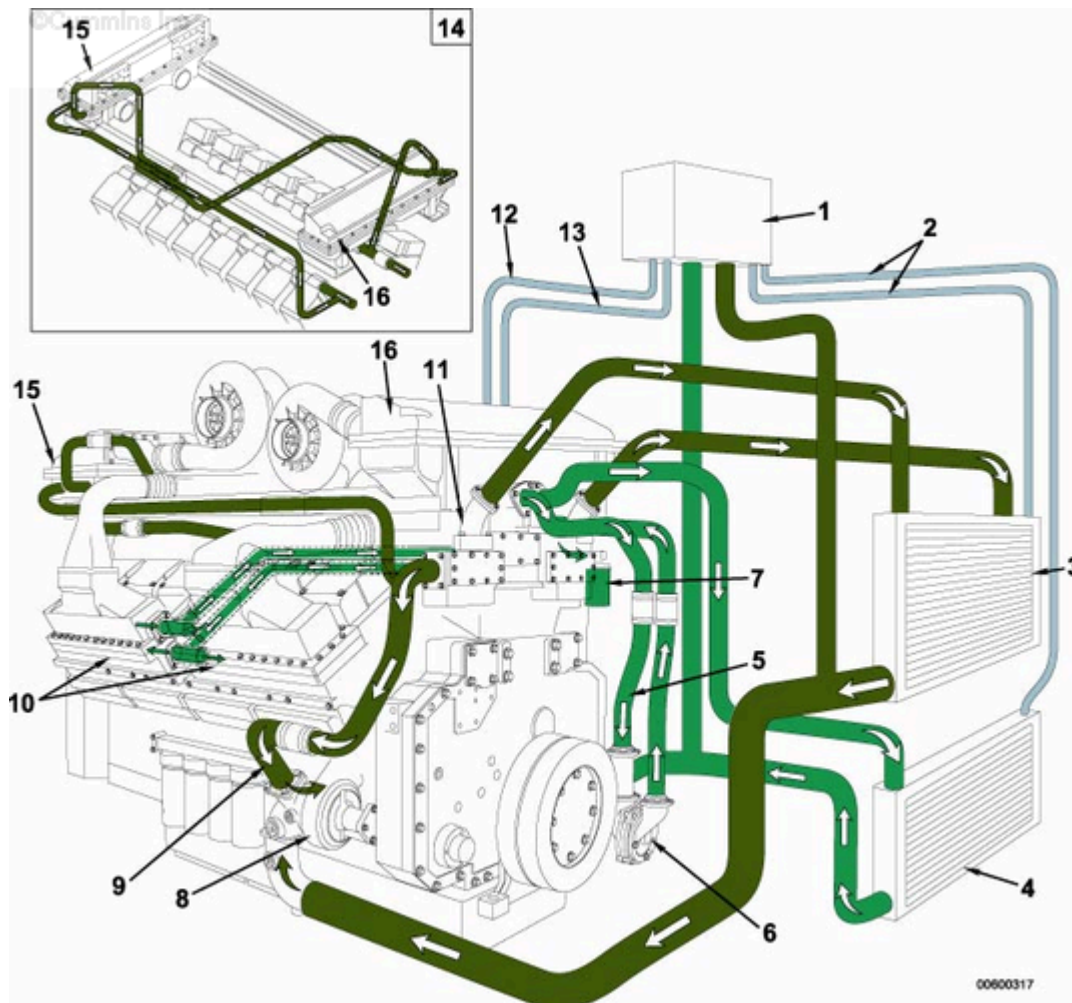
QSK45 Cooling System

1. Surge tank
2. Radiator vents
3. Engine radiator
4. LTA radiator
5. Low temperature bypass (from thermostat housing)
6. LTA water pump
7. Coolant filters
8. Engine water pump
9. Engine coolant bypass (from thermostat housing)
10. Aftercoolers
11. Thermostat housings
12. Engine coolant vent (from thermostat housing)
13. Low temperature coolant vent (from thermostat housing).



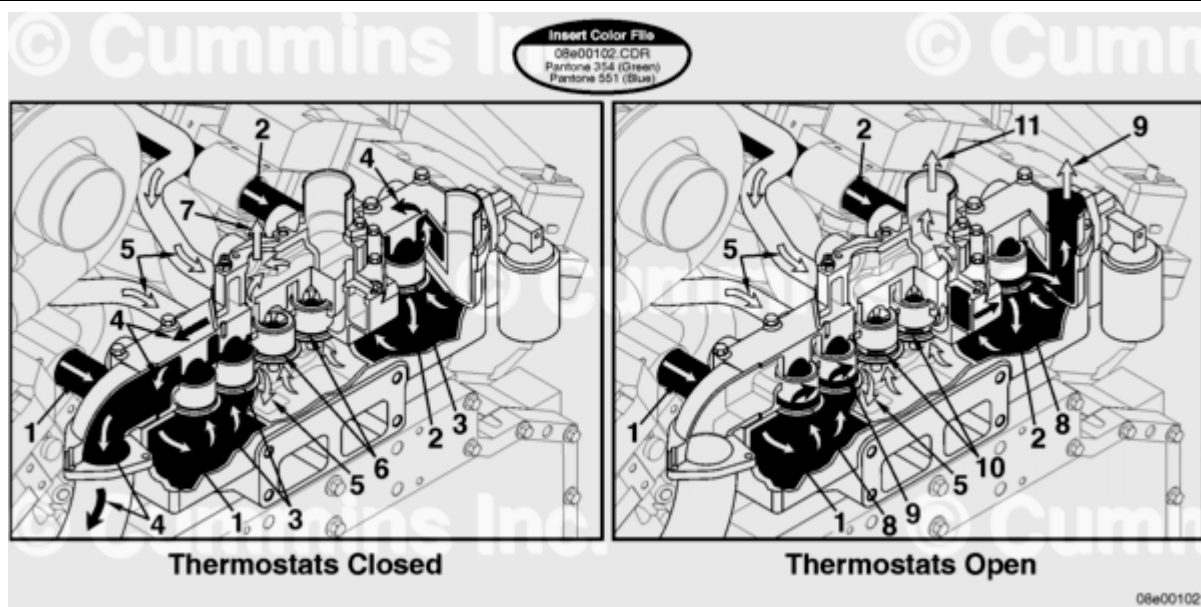
QSK60 Cooling System - Single-Stage Turbocharged, Aftercooled

1. Surge tank
2. Radiator vents
3. Engine radiator
4. LTA radiator
5. Low temperature bypass (from thermostat housing)
6. LTA water pump
7. Coolant filters
8. Engine water pump
9. Engine coolant bypass (from thermostat housing)
10. Aftercoolers
11. Thermostat housings
12. Engine coolant vent (from thermostat housing)
13. Low temperature coolant vent (from thermostat housing).



QSK60 Cooling System - Two-Stage Turbocharged, Aftercooled

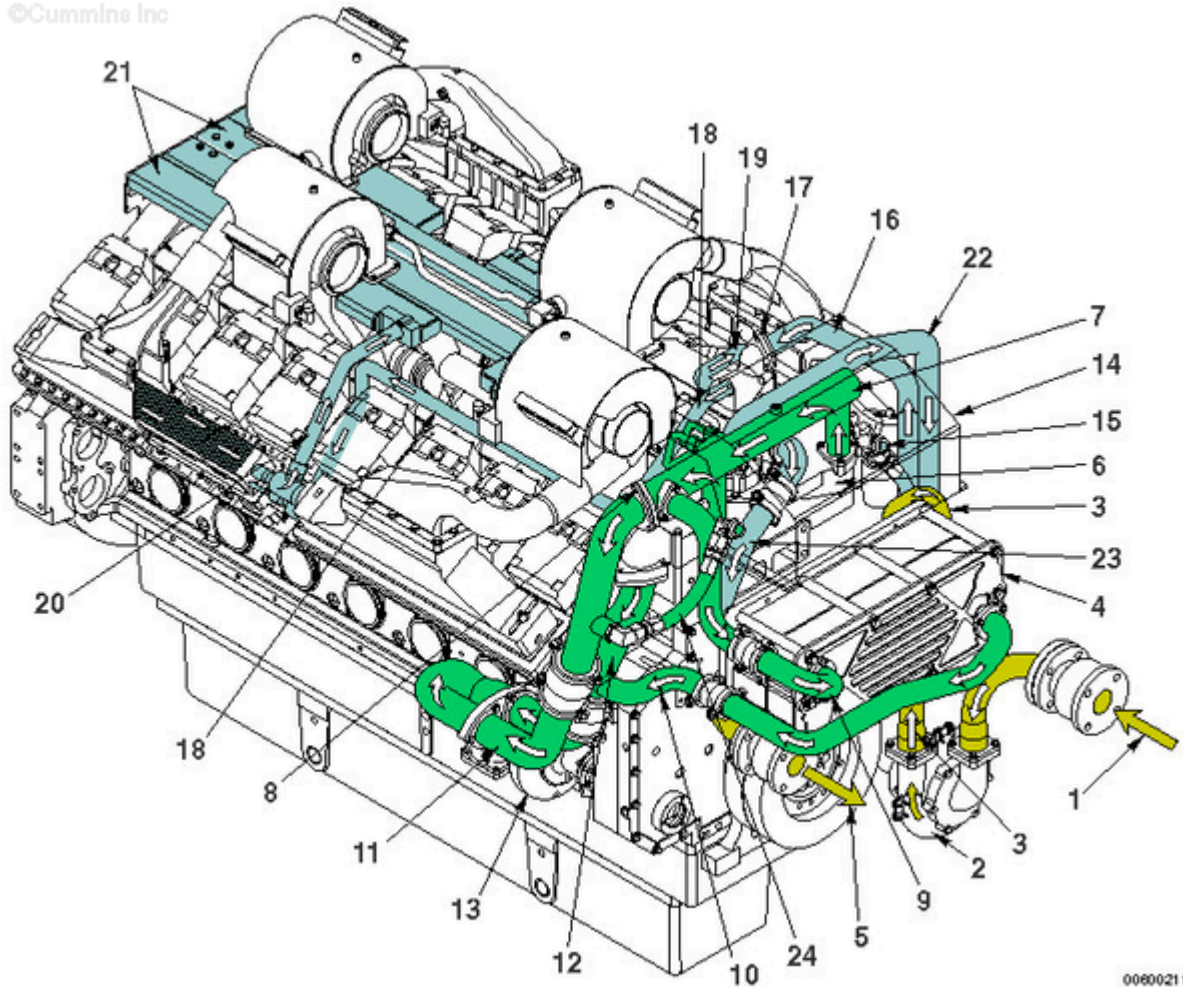
1. Surge tank
2. Radiator vents
3. Engine radiator
4. LTA radiator
5. Low temperature bypass (from thermostat housing)
6. LTA water pump
7. Coolant filters
8. Engine water pump
9. Engine coolant bypass (from thermostat housing)
10. Stage two aftercoolers (low temperature circuit)
11. Thermostat housings
12. Engine coolant vent (from thermostat housing)
13. Low temperature coolant vent (from thermostat housing)
14. Inset - showing stage one aftercooling
15. Stage one intercoolers (rear)
16. Stage one intercoolers (front).



Thermostat Housing Flow - Low temperature aftercooler (two pump/two loop), with Mechanically Actuated Injectors

1. Right bank engine coolant to thermostat housing
2. Left bank engine coolant to thermostat housing
3. Engine thermostats (closed)
4. Engine coolant bypass to engine water pump
5. LTA coolant supply to thermostat housing
6. LTA thermostats (closed)
7. LTA coolant bypass to low temperature aftercooler water pump
8. Engine thermostats (open)
9. Engine coolant to engine radiator
10. LTA thermostats (open)
11. LTA coolant return to low temperature aftercooler radiator.

Note : If the thermostats are partially open, coolant is flowing through both the radiator and the bypass.

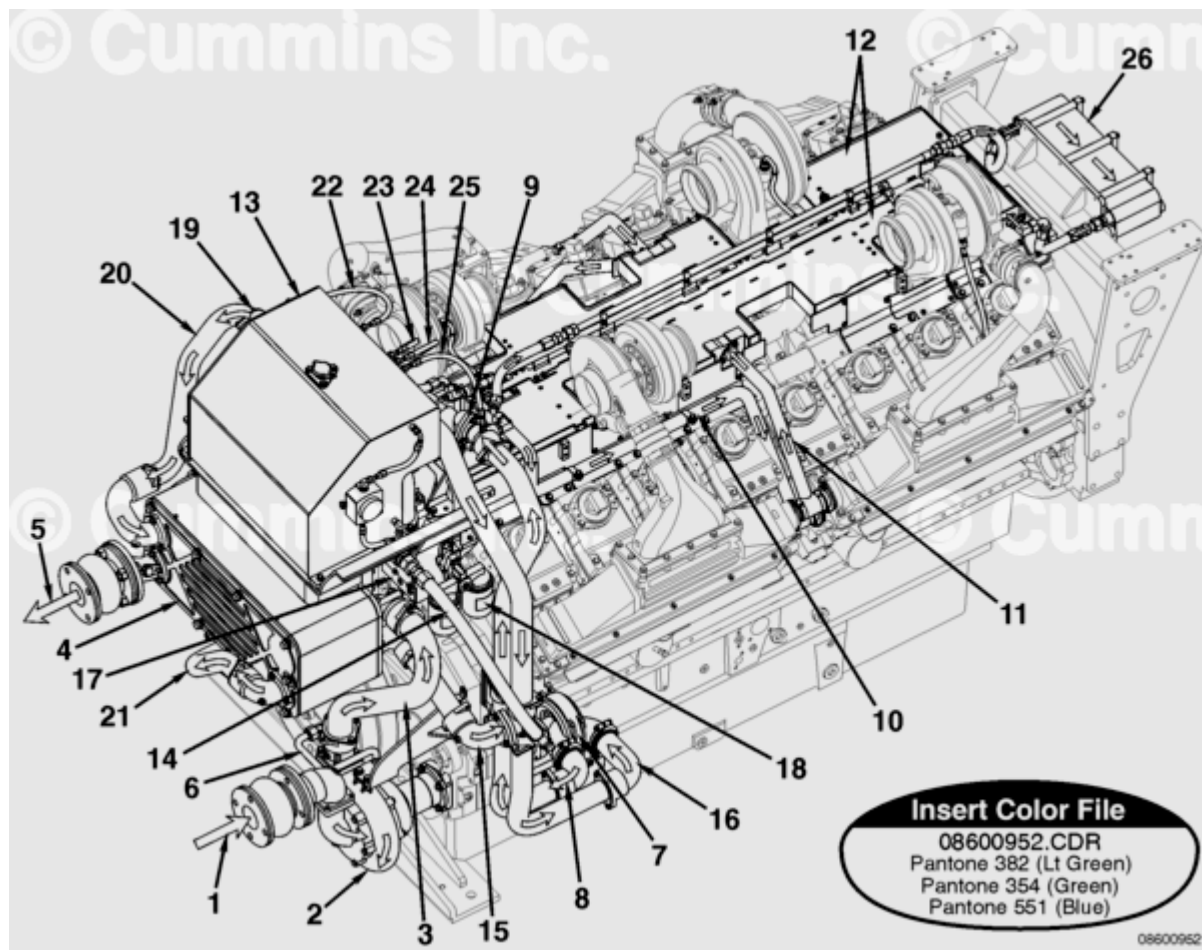


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QSK60 with Heat Exchanger - Marine

1. Sea water inlet
2. Sea water pump
3. Sea water pump outlet
4. Heat exchanger
5. Sea water outlet from heat exchanger
6. Thermostat housing
7. Engine coolant outlet manifold
8. Engine coolant bypass orifice plate
9. Engine coolant supply to heat exchanger
10. Heat exchanger engine coolant return to engine water pump
11. Engine coolant heat exchanger bypass to engine water pump
12. Engine coolant thermostat bypass to engine water pump
13. Engine water pump
14. Expansion tank
15. LTA coolant make-up from expansion tank
16. LTA water pump outlet
17. "J" block
18. LTA coolant supply to right bank aftercoolers
19. LTA coolant supply to left bank aftercoolers.
20. LTA coolant supply to right bank exhaust manifold heat shields
21. Exhaust manifold heat shields

- 22. LTA thermostat bypass to low temperature aftercooler water pump
- 23. LTA coolant supply to heat exchanger
- 24. Engine coolant make-up from expansion tank.

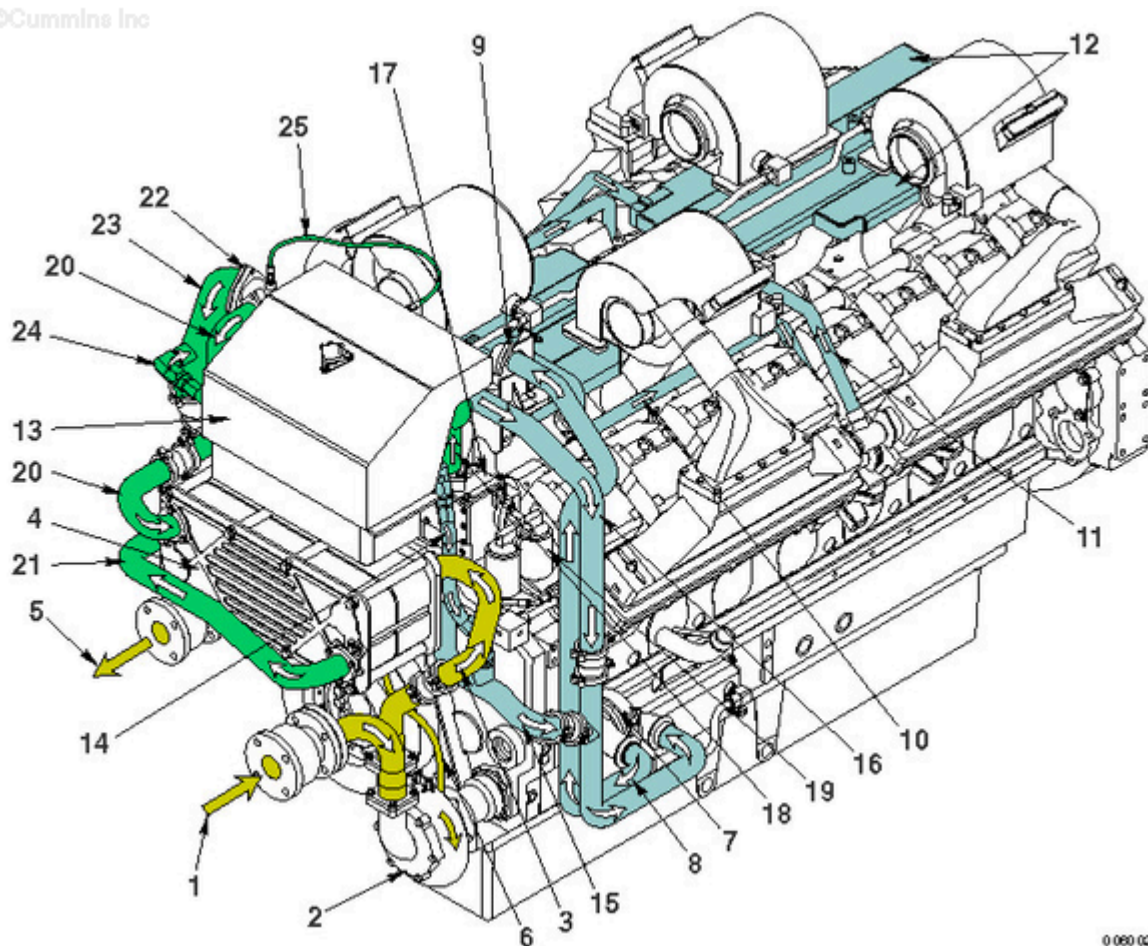


QSK60 Tier 2 with Heat Exchanger - Marine

- 1. Sea water inlet
- 2. Sea water pump
- 3. Sea water inlet to heat exchanger
- 4. Heat exchanger
- 5. Sea water outlet from heat exchanger
- 6. Sea water pump priming impeller discharge line
- 7. LTA coolant pump
- 8. LTA coolant pump
- 9. "J" block
- 10. LTA coolant supply to left back aftercoolers
- 11. LTA coolant supply to left bank exhaust manifold heat shields
- 12. Exhaust manifold heat shields
- 13. Expansion tank
- 14. LTA coolant make-up expansion tank
- 15. Heat exchanger LTA coolant outlet to LTA coolant pump
- 16. LTA thermostat bypass to LTA coolant pump
- 17. Thermostat housing
- 18. Engine coolant filters

19. Engine coolant bypass orifice plate
20. Engine coolant supply to heat exchanger
21. Engine coolant return to engine coolant pump
22. Engine coolant manifold vent line
23. Turbocharger heat shield vent line
24. Exhaust manifold heat shield vent line
25. Engine coolant manifold vent line
26. Lubricating oil cooler.

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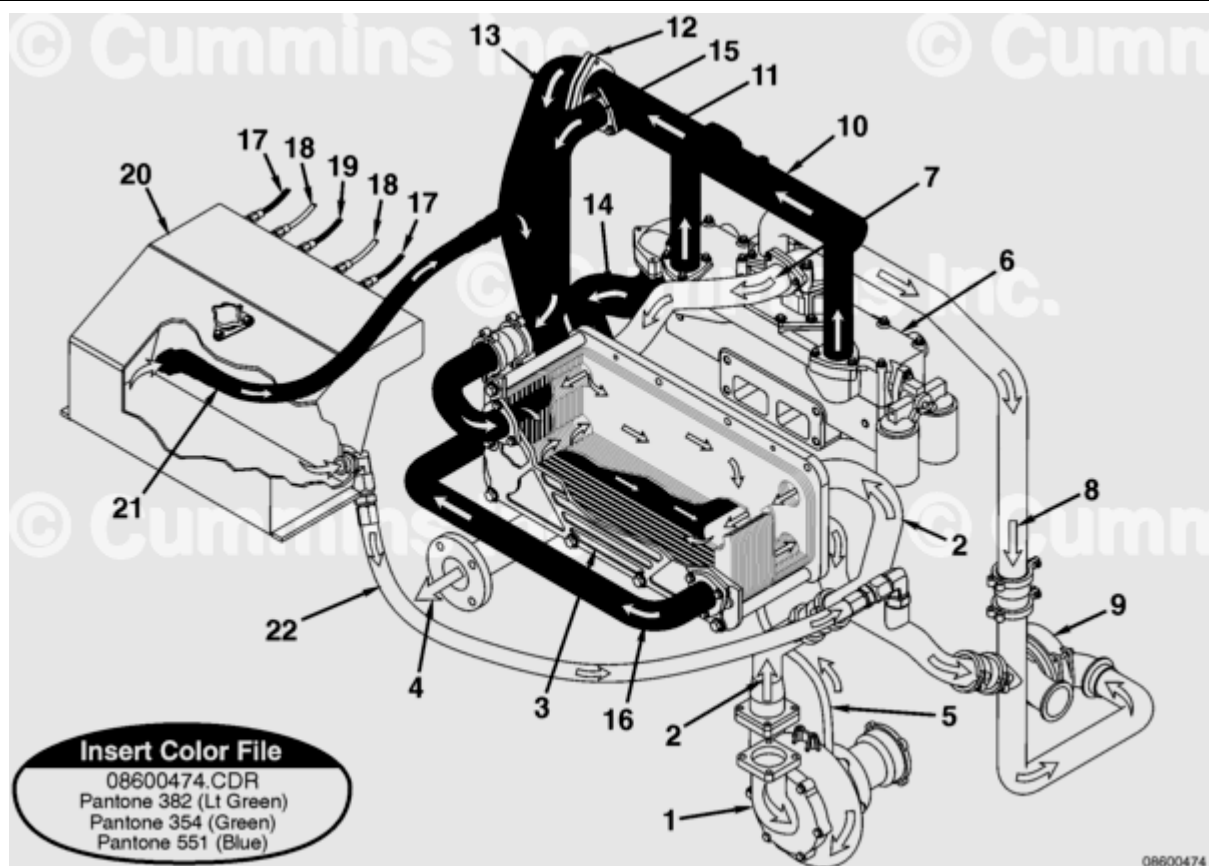


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QSK60 with Heat Exchanger - Marine

1. Sea water inlet
2. Sea water pump
3. Sea water pump outlet to heat exchanger
4. Heat exchanger
5. Sea water outlet from heat exchanger
6. Sea water pump priming impeller discharge line
7. LTA water pump
8. LTA water pump outlet
9. "J" block
10. LTA coolant supply to left-bank aftercoolers
11. LTA coolant supply to left-bank exhaust manifold heat shields
12. Exhaust manifold heat shields
13. Expansion tank

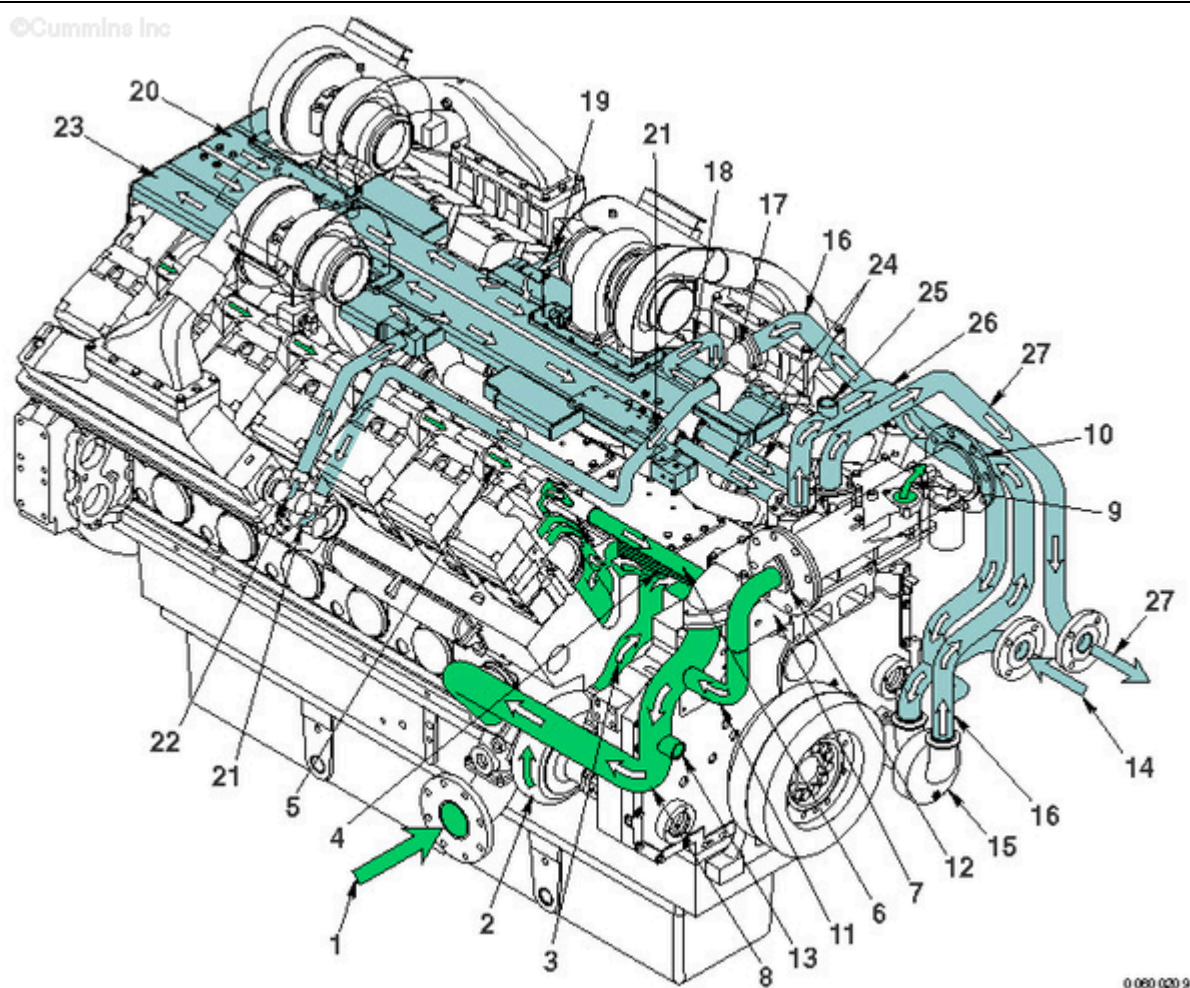
14. LTA coolant make-up from expansion tank
15. Heat exchanger low temperature aftercooler coolant outlet to low temperature aftercooler water pump
16. LTA thermostat bypass to low temperature aftercooler water pump
17. Thermostat housing
18. Engine coolant outlet manifold
19. Engine coolant filters
20. Engine coolant supply to heat exchanger
21. Engine coolant return to engine water pump
22. Engine coolant bypass orifice plate
23. Engine coolant heat exchanger bypass to engine water pump
24. Engine coolant make-up from expansion tank
25. Engine coolant manifold vent line.



QSK60 Heat Exchanger - Marine

1. Sea water pump
2. Sea water pump outlet to heat exchanger
3. Heat exchanger
4. Sea water outlet from heat exchanger
5. Sea water pump priming impeller discharge line
6. Thermostat housing
7. LTA coolant supply to heat exchanger
8. LTA bypass to low temperature aftercooler water pump
9. LTA water pump

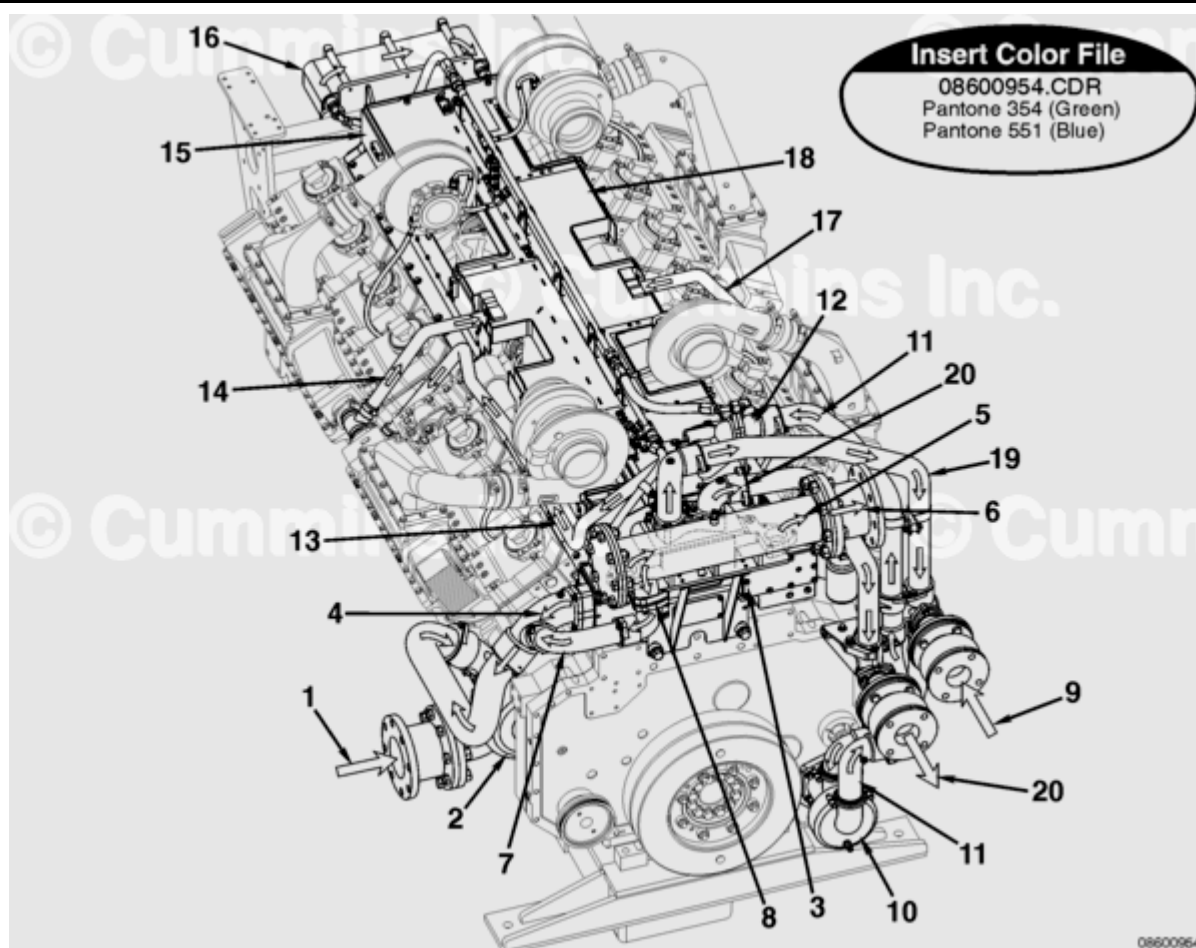
10. Engine coolant outlet manifold
11. Engine coolant outlet from thermostat housing
12. Engine coolant bypass orifice plate
13. Engine coolant heat exchanger bypass to engine water pump
14. Engine coolant thermostat bypass to engine water pump
15. Engine coolant supply to heat exchanger
16. Heat exchanger engine coolant return to engine water pump
17. Turbocharger heat shield vent line
18. Exhaust manifold heat shield vent line
19. Engine coolant manifold vent line
20. Expansion tank
21. Engine coolant make-up from expansion tank
22. LTA coolant make-up from expansion tank.



QSK60 Keeled Cooled - Marine

1. Engine coolant inlet from main engine cooler
2. Engine water pump
3. Engine coolant supply to block
4. Lubricating oil cooler
5. Engine coolant manifold
6. Engine coolant to thermostat housing
7. Thermostat housing

8. Engine coolant thermostat bypass to engine water pump
9. Engine coolant to engine coolant outlet manifold
10. Engine coolant outlet to main engine cooler
11. Main engine cooler bypass to engine water pump
12. Engine coolant bypass orifice plate
13. Engine coolant make-up from expansion tank
14. LTA coolant inlet from LTA cooler
15. LTA water pump
16. LTA water pump outlet to "J" block
17. "J" block
18. LTA coolant from "J" block to left bank aftercoolers
19. LTA coolant from left-bank aftercoolers to left bank exhaust manifold heat shields
20. Left bank exhaust manifold heat shields
21. LTA coolant from "J" block to right bank aftercoolers
22. LTA coolant from right-bank aftercoolers to right bank exhaust manifold heat shields
23. Right bank exhaust manifold heat shield
24. LTA coolant from exhaust manifold heat shields to thermostat housing
25. LTA coolant make-up from expansion tank
26. LTA coolant thermostat bypass to LTA water pump
27. LTA coolant outlet to LTA cooler.

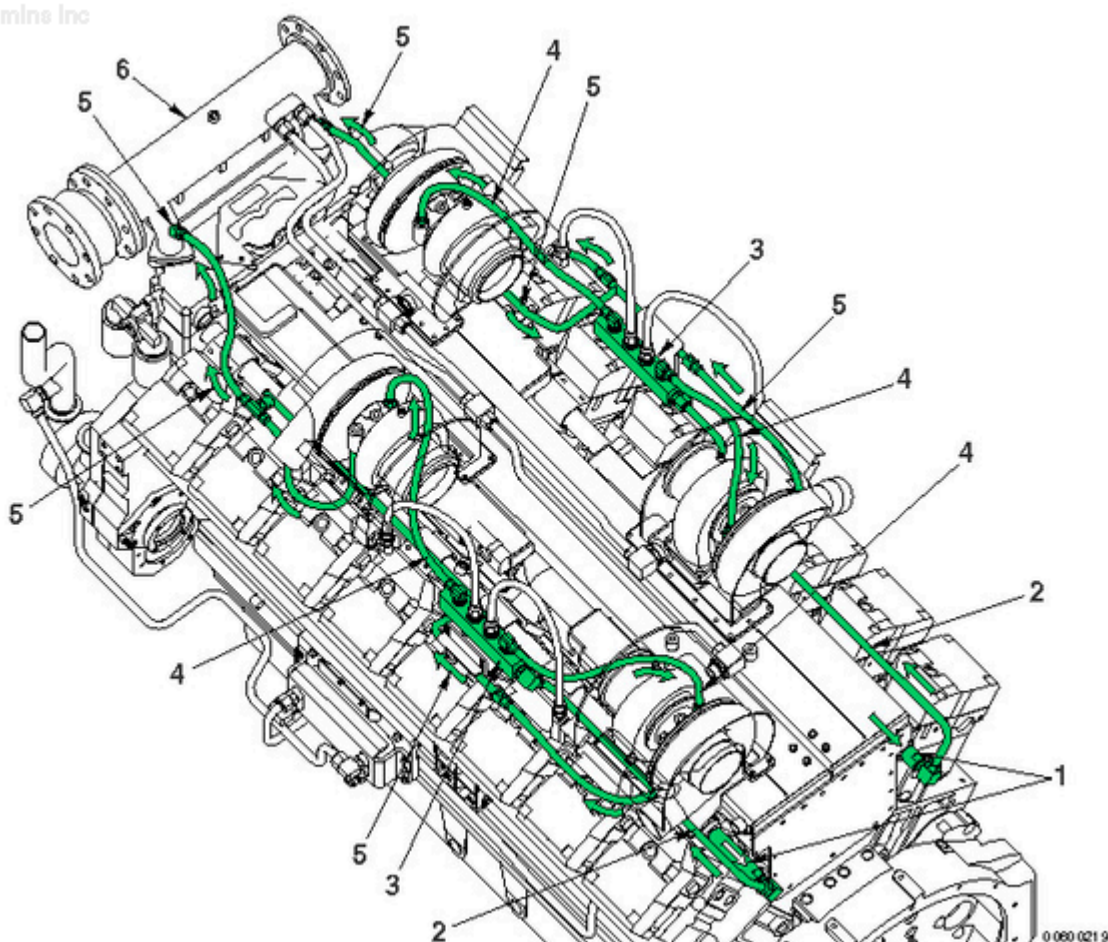


QSK60 Tier 2 Keel Cooled - Marine

1. Engine coolant inlet from main engine cooler

2. Engine water pump
3. Thermostat housing
4. Engine coolant thermostat bypass to engine water pump
5. Engine coolant to engine coolant outlet manifold
6. Engine coolant outlet to main engine cooler
7. Main engine cooler bypass to engine water pump
8. Engine coolant bypass orifice plate
9. LTA coolant inlet from low temperature aftercooler cooler
10. LTA water pump
11. LTA water pump outlet to "J" block
12. "J" block
13. LTA coolant from "J" block to right bank aftercoolers
14. LTA coolant from right bank aftercoolers to right bank exhaust manifold heat shields
15. Right bank exhaust manifold heat shields
16. Lubricating oil cooler
17. Left bank aftercoolers to left bank exhaust manifold heat shields
18. Left bank exhaust manifold heat shields
19. LTA coolant thermostat bypass to LTA water pump
20. LTA coolant outlet to LTA cooler.

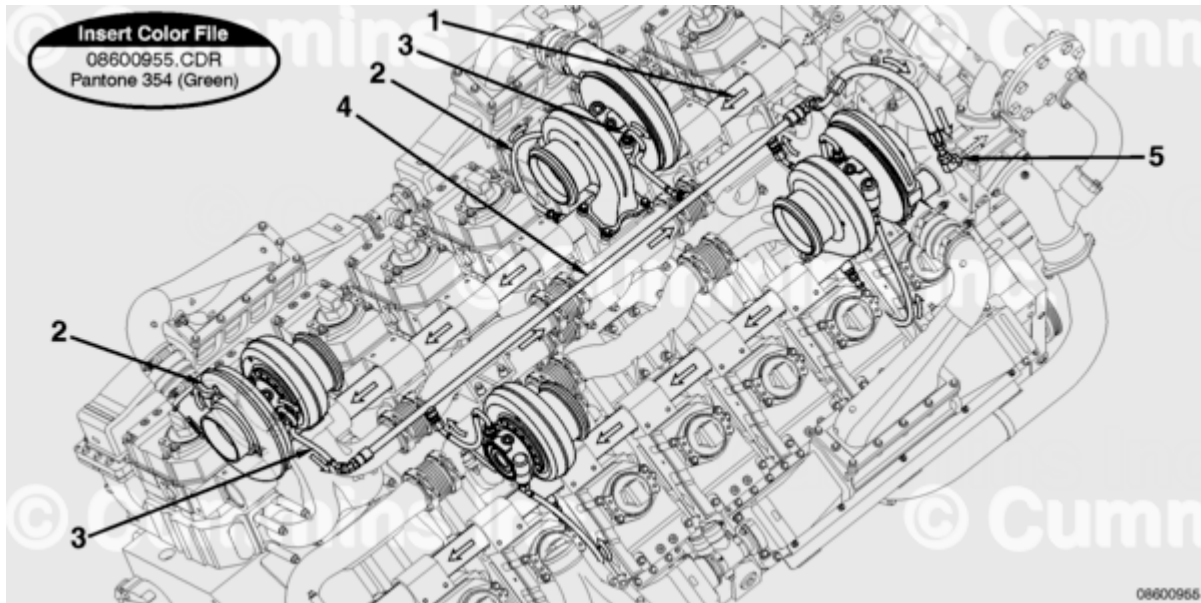
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QSK60 Turbocharger Bearing Housing Cooling - Marine

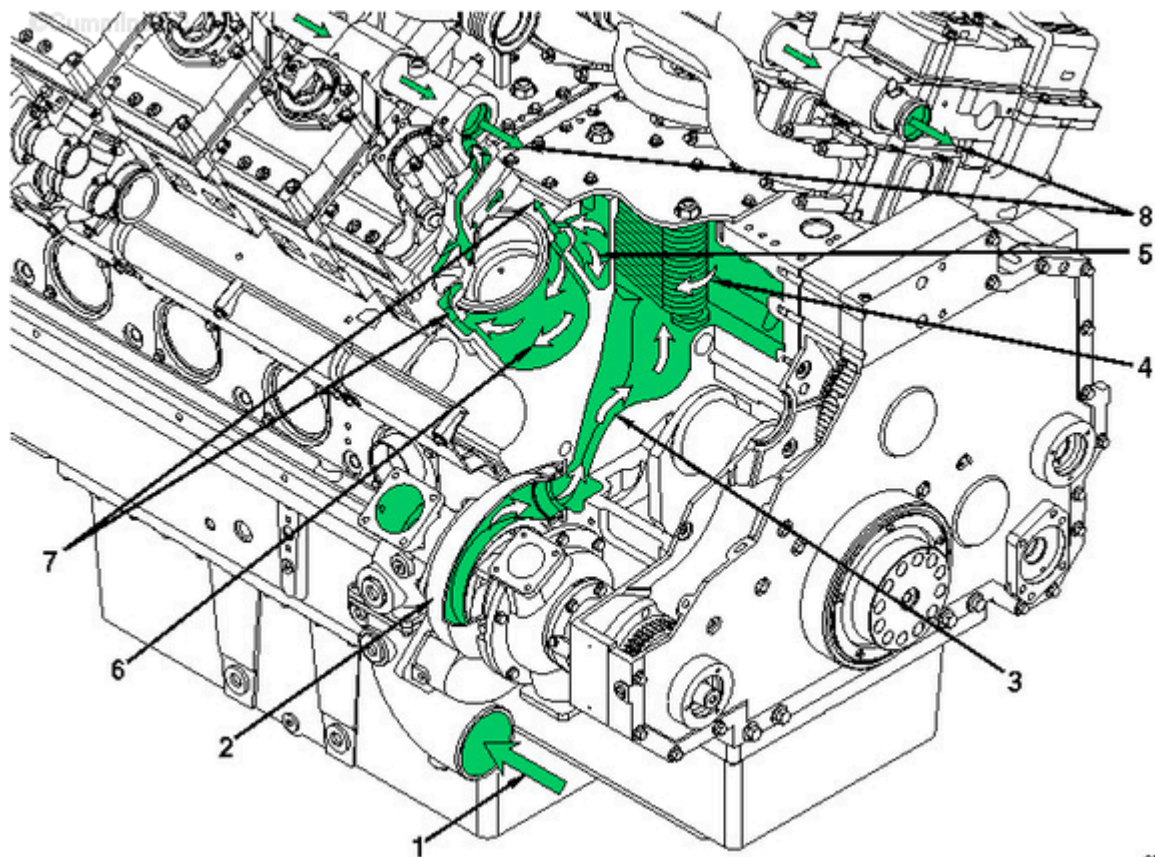
1. Engine coolant from engine coolant manifold
2. Engine coolant to engine coolant distribution block

3. Engine coolant distribution block
4. Engine coolant supply to turbocharger bearing housing
5. Engine coolant return from turbocharger bearing housing
6. Engine coolant outlet manifold.



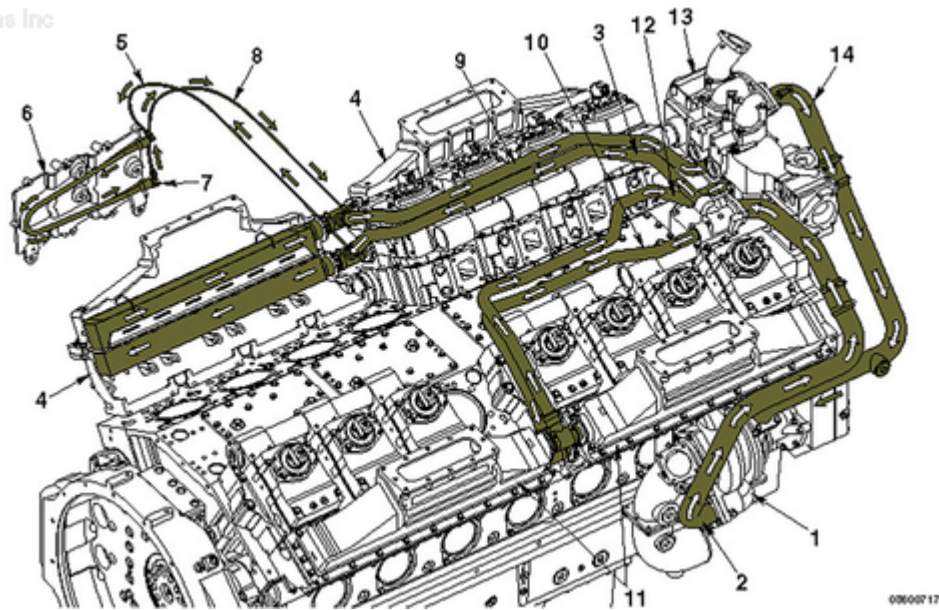
QSK60 Tier 2 Turbocharger Bearing Housing Cooling - Marine

1. Engine coolant flow from thermostat housing
2. Engine coolant inlet to turbocharger bearing housing
3. Engine coolant outlet from turbocharger bearing housing to manifold tube
4. Engine coolant manifold tube
5. Engine coolant return to thermostat housing.



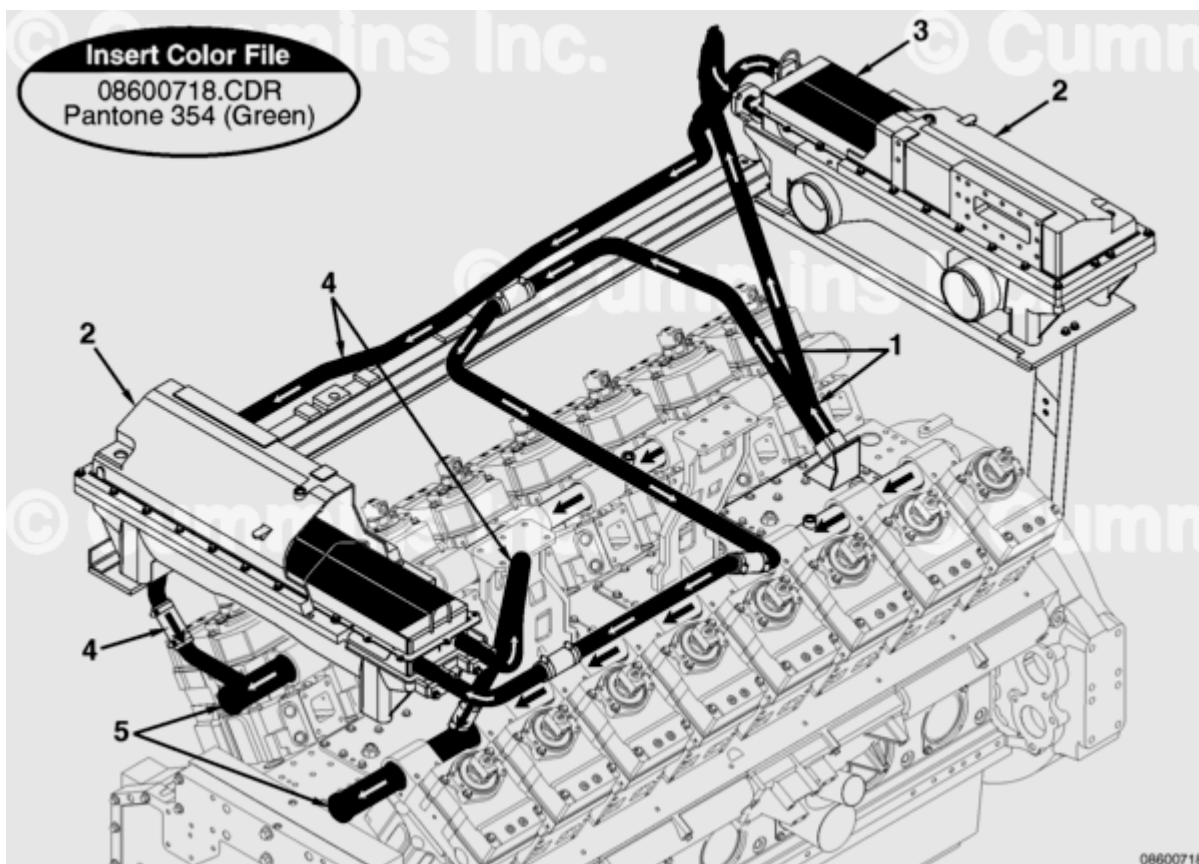
QSK60 Jacket Water Coolant Flow - with Electronically Actuated Injectors

1. Coolant inlet
2. Water pump
3. Coolant flow to oil cooler cavity
4. Coolant around oil coolers
5. Water jacket coolant
6. Coolant flow around cylinder liner
7. Coolant flow around cylinder head
8. Coolant flow to thermostat housing.



QSK60 Low Temperature Aftercooler Flow - with Electronically Actuated Injectors

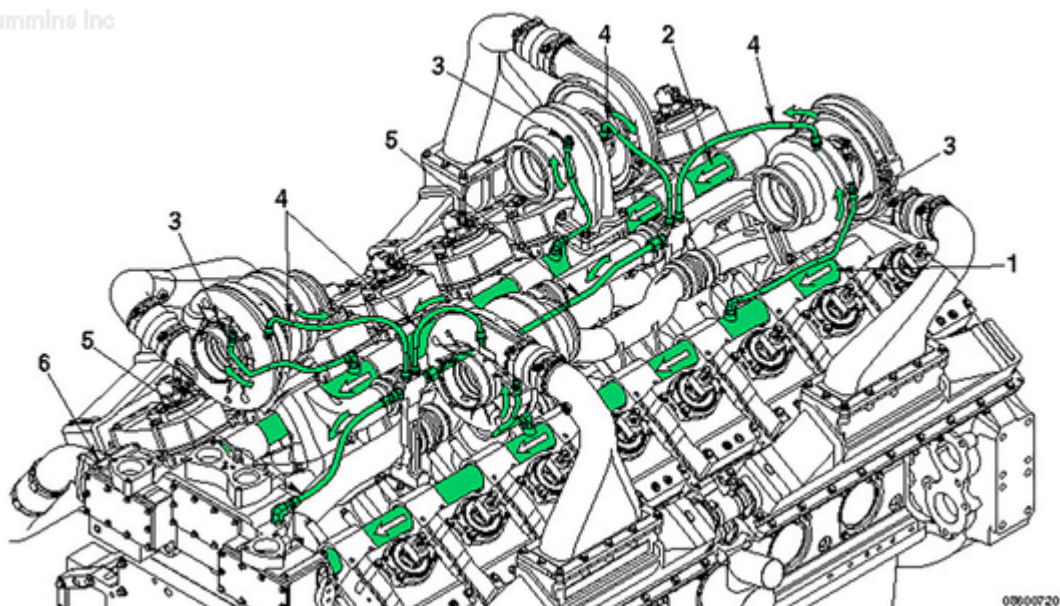
1. LTA coolant pump
2. Coolant supply
3. Left bank aftercooler coolant supply
4. Left bank aftercooler
5. Coolant supply to electronic control module (ECM) coolant plate
6. Electronic control module (ECM) coolant plate
7. Coolant drain valve
8. Electronic control module (ECM) coolant plate coolant return
9. Left bank aftercooler coolant return
10. Right bank aftercooler coolant supply
11. Right bank aftercooler
12. Right bank aftercooler coolant return
13. Thermostat housing
14. LTA coolant bypass.



QSK60 Intercooler Coolant Flow - with Electronically Actuated Injectors

1. Coolant supply from oil coolers
2. Intercooler
3. Intercooler core
4. Coolant return to water manifold
5. Coolant return to thermostat housing.

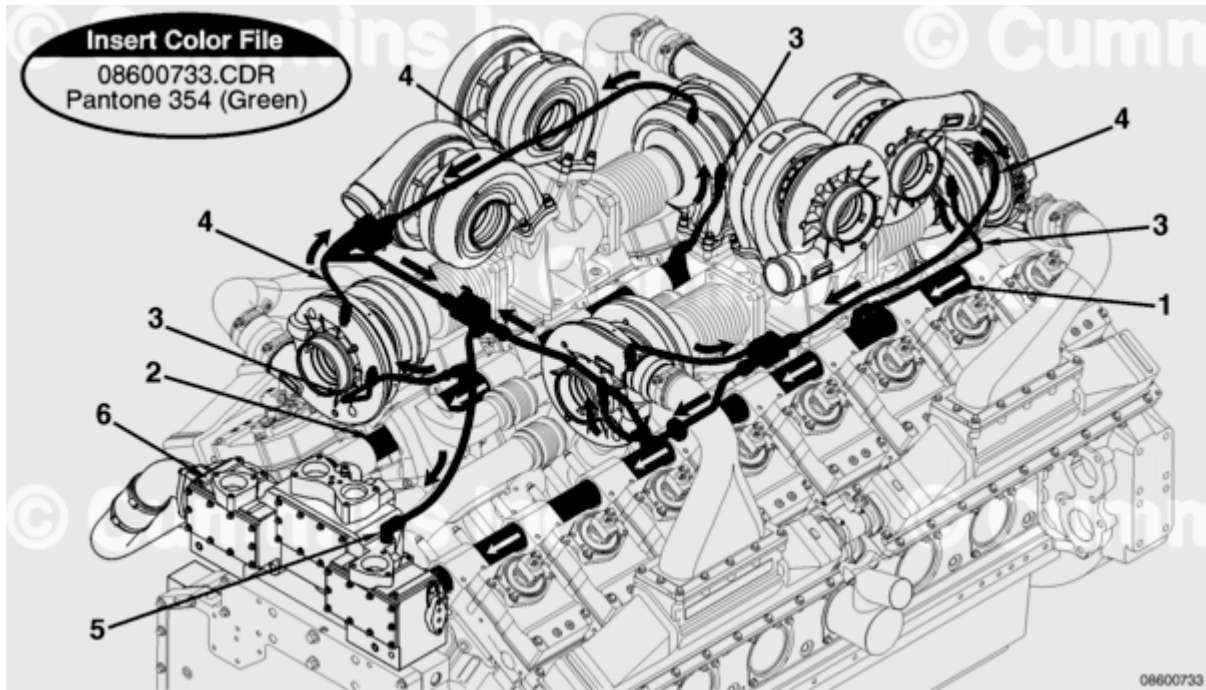
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QSK60 Single-Stage Turbocharger Coolant Flow - with Electronically Actuated Injectors

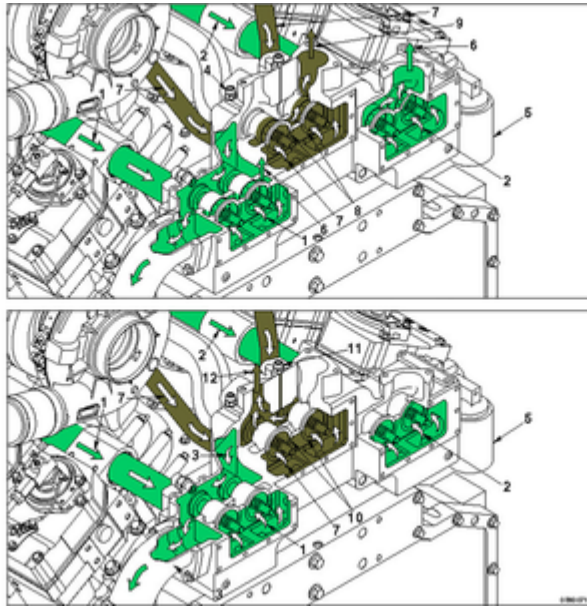
1. Left bank water manifold
2. Right bank water manifold

3. Turbocharger coolant supply
4. Turbocharger coolant return
5. Turbocharger coolant return to thermostat housing
6. Thermostat housing.



QSK60 Two-Stage Turbocharger Coolant Flow - with Electronically Actuated Injectors

1. Left bank water manifold
2. Right bank water manifold
3. Turbocharger coolant supply
4. Turbocharger coolant return
5. Turbocharger coolant return to thermostat housing
6. Thermostat housing.



QS60 Thermostat Housing Coolant Flow (top view open, bottom view closed) - with
Electronically Actuated Injectors

1. Right bank coolant to thermostat housing
2. Left bank coolant to thermostat housing
3. Coolant bypass to water jacket pump
4. Coolant vent port
5. Coolant filters
6. Coolant return to radiators
7. LTA coolant from aftercooler
8. LTA thermostats open
9. LTA coolant supply to LTA radiator
10. LTA thermostats closed
11. LTA vent
12. LTA coolant bypass to LTA pump.